Divide quantities in a given ratio

1 James and Emma scored goals in their netball game in the ratio $4 : 3$.

- **a** Find the fraction of the goals scored by James.
- **b** If they scored a total of 56 goals, how many goals did James score?
- **c** How many goals did Emma score?

2 Harris and Bella divided lollies between them in the ratio $5 : 4$.

- **a** Find the fraction of lollies that Bella gets.
- **b** If there are 27 lollies, how many does Bella get?
- **c** How many lollies does Harris get?

3 56 building blocks are shared between Mohamad and Isabelle in the ratio $2 : 5$.

- **a** Find the fraction of the blocks Mohamad receive.
- **b** How many blocks does Mohamad receive?
- **c** How many blocks does Isabelle receive?

4 Ben always buys melons and bananas in the ratio $7 : 3$. If he buys 30 pieces of fruit in total, how many melons did he buy?

5 The ratio of adults to children on a train is $2 : 7$. The train is carrying 360 passengers.

- **a** How many adults are on the train?
- **b** How many children are on the train?

6 James and Sophia share $\$880$ in the ratio $2 : 9$.

- **a** How much money does James get?
- **b** How much money does Sophia get?

7 Avril is making a scarf that uses two materials. She wants to use the materials in the ratio of $\frac{11}{15}$ metres of red for every $\frac{3}{5}$ metres of green.

- **a** Write the simplified ratio of red material to green material.
- **b** If she needs 33 metres of red material, how many metres of green material will she need?

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8 Neil is mixing paint to get a specific shade of orange. He wants a mix of red and yellow paint in the ratio of $3\frac{3}{4} : 6\frac{2}{3}$.

a Simplify the ratio.

b How many litres of red paint will Neil need if he wants to make 75 litres of orange paint?

9 Kathleen and Luke invest money in a business in the ratio $3 : 10$.

a What fraction did Kathleen contribute?

b If Kathleen has invested $\$2100$, how much did Luke invest?

10 At a sporting event, the ratio of security personnel to spectators is $1 : 180$. If $34\,200$ spectators attend the event, how many security personnel will be required?

11 Two kinds of pine trees, type A and type B, are planted in rows. In each row the ratio of type A to type B is $9 : 8$ Altogether, 180 type A trees are planted.

How many type B trees are planted?

12 Xavier and Ursula split the cost of the groceries in the ratio $0.5 : 1.5$. Altogether the groceries cost $\$140$.

a Simplify the ratio.

b How much did Xavier pay for the groceries?

13 A car and a truck are driving down the highway. Their relative speeds are in the ratio $3.1 : 2.9$. If the car is travelling at $93\ \text{km/hr}$, how fast is the truck travelling?

14 In a zoo the ratio of elephants to lions is $7 : 4$.

a Find the ratio of elephants to lions to the total number of elephants and lions.

b Complete the following table of equivalent ratios:

Elephants	to	Lions	to	Total
7	:	4	:	
14	:		:	
	:		:	110

c If there are 66 elephants and lions altogether, how many lions are there?

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15 The ratio of red, green and black coloured area on a dart board is $12 : 7 : 31$. The area of the dart board is $150\ \text{cm}^2$.

a Find the number of ratio parts corresponding to the following:

i Red area **ii** Green area **iii** Black area **iv** Total area

b Find the size of a single ratio part.

c Find the area of the following:

i Red area **ii** Green area **iii** Black area

16 Vanessa competed in a marathon and found that her times for the swimming, cycling and running parts were in the ratio $3 : 1 : 2$. If the quickest part took 10 minutes, how many minutes did she take to complete the whole marathon?

17 The ratio of students who play badminton, football and softball is $2 : 5 : 9$.

a If there are 72 students who play softball, how many students play badminton?

b Find the total number of students who play one of these sports.

18 The perimeter of a rectangle is $42\ \text{cm}$ and the ratio of its length to its width is $4 : 3$.

a How many parts are in the ratio?

b Find the length of the rectangle.

c Find the width of the rectangle.

Increase and decrease in a given ratio (Extended)

19 Increase the following quantities in the ratio specified:

a $\$400$ in the ratio $3 : 1$.

b $\$780$ in the ratio $5 : 2$.

c $60\ \text{kg}$ in the ratio $4 : 3$.

d $800\ \text{g}$ in the ratio $3 : 2$.

e $1.5\ \text{L}$ in the ratio $2 : 1$.

f $1200\ \text{mL}$ in the ratio $6 : 5$.

g $120\ \text{cm}$ in the ratio $10 : 2$.

h $5.2\ \text{m}$ in the ratio $100 : 5$.

20 Decrease the following quantities in the ratio specified:

a $\$500$ in the ratio $1 : 5$.

b $\$820$ in the ratio $3 : 4$.

c $50\ \text{kg}$ in the ratio $2 : 5$.

d $150\ \text{g}$ in the ratio $1 : 2$.

e $2.4\ \text{L}$ in the ratio $5 : 6$.

f $900\ \text{mL}$ in the ratio $4 : 5$.

g $30\ \text{cm}$ in the ratio $8 : 10$.

h $3.4\ \text{m}$ in the ratio $5 : 20$.

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21 A clothing store is having a sale and is decreasing the price of all their items in the ratio $3 : 4$. Find the new sale price of the following items:

a A jumper originally priced at $\$50$.

b A dress originally priced at $\$120$.

c A pair of jeans originally priced at $\$80$.

d A skirt originally priced at $\$75$.

22 At the beginning of the year, an office supplies store increases the prices of its items in the ratio of $7 : 6$. Find the new increased prices of the following items:

a A pencil case originally priced at $\$12$.

b A desk originally priced at $\$180$.

c A pair of jeans originally priced at $\$54$.

d A set of shelves originally priced at $\$150$.

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